

Path 1: Pre Security > Module 6: Software Basics > Room 3: Python Simple Demo

<https://tryhackme.com/room/pythonsimpledemo>

Python: Simple Demo

Explore what a basic Python program looks like.

>> Task 1: Introduction

You may have heard of the Python programming language numerous times. Python is a high-level general-purpose programming language. By high-level, we mean it hides most implementation details; general-purpose means you can use Python for a wide variety of scenarios, from web applications and automation scripts to data science and machine learning.

>> Task 2: Variables

```
import random # gives us tools for picking random numbers

secret = random.randint(1, 20) # a <= secret <= b
tries = 0
guess = 0 # start with a value that cannot be the secret (since secret is 1..20)

print("I'm thinking of a number between 1 and 20")

text = input("Take a guess: ") # input() returns text (a string)
guess = int(text) # convert the text to a number

tries = tries + 1 # add 1 try
```

Q1) What is the name of the function we used to display text on the screen?

Answer: print()

Q2) What is the name of the function that we used to convert user input to an integer?

Answer: int()

>> Task 3: Conditional Statements

```
import random # gives us tools for picking random numbers

secret = random.randint(1, 20) # a <= secret <= b
tries = 0
guess = 0 # start with a value that cannot be the secret (since secret is 1..20)

print("I'm thinking of a number between 1 and 20")

text = input("Take a guess: ") # input() returns text (a string)
guess = int(text) # convert the text to a number

tries = tries + 1 # add 1 try

# Give a hint using if / elif / else.
if guess < 1 or guess > 20:
    print("That number is out of range. Try again.")
elif guess < secret:
    print("Too low, try again.")
elif guess > secret:
    print("Too high, try again.")
else:
    print("You got it in", tries, "tries!")
```

Q1) How does Python write “else if”?

Answer: elif

Q2) What will the program display if the user’s input is 50?

Answer: That number is out of range. Try again.

>> Task 4: Iterations

```
import random # gives us tools for picking random numbers

# -----
# Guess the Number (Beginner Demo)
# -----
# The computer picks a secret number.
# The player keeps guessing until they find it.

secret = random.randint(1, 20) # a <= secret <= b
tries = 0
guess = 0 # start with a value that cannot be the secret (since secret is 1..20)

print("I'm thinking of a number between 1 and 20")

# Repeat until the user guesses the secret number.
while guess != secret:
    text = input("Take a guess: ") # input() returns text (a string)
    guess = int(text) # convert the text to a number

    tries = tries + 1 # add 1 try

    # Give a hint using if / elif / else.
    if guess < 1 or guess > 20:
        print("That number is out of range. Try again.")
    elif guess < secret:
        print("Too low, try again.")
    elif guess > secret:
        print("Too high, try again.")
    else:
        print("You got it in", tries, "tries!")
```

Q1) What type of loop does this program use?

Answer: while

Q2) What will the program display if the user makes the correct guess in 3 tries?

Answer: You got it in 3 tries!